ADM/ADP32F036 系列 BOOT/APP 程序核查与修改指南

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技术文件修订记录

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1 BOOT 程序

1.1 更改 Flash 等待时间

100M 主频，Flash 等待时间 3，3，5 改成 4，4，5。

HardwareSetup.c

```
53 Date: 2025-11-07
54 Parameters: None
55 Returns: None
56 ************************************************************
57 void Flash_Initialize(void)
58 {
59     EALLOW;  // Enable writes to protected registers
60
61     // Enable Flash Pipeline mode to improve performance o
62     FlashRegs.FOPT.bit.ENPIPE = 1;
63
64     // Set the Paged Waitstate for the Flash
65     FlashRegs.FBANKWAIT.bit.PAGEWAIT = 3;
66
67     // Set the Random Waitstate for the Flash
68     FlashRegs.FBANKWAIT.bit.RANDWAIT = 3;
69
70     // Set the Waitstate for the OTP (One-Time Programmabl
71     FlashRegs.FOTPWAIT.bit.OTPWAIT = 5;
72
73     // CAUTION: ONLY USE DEFAULT VALUES FOR THESE 2 REGIST
74     FlashRegs.FSTDBYWAIT.bit.STDBYWAIT = 0x01FF;   // Stan
75     FlashRegs.FACTIVEWAIT.bit.ACTIVEWAIT = 0x01FF; // Acti
76
77     EDIS;  // Disable writes to protected registers
78
79     // Force a pipeline flush to ensure the write to the l
80     asm(" RPT #7 || NOP");  // 7 NOP operations for pipeli
81 }
```

1.2 更改中断向量

跳过地址 0xD00~0xD05，0xD00~xD05 不属于中断向量，添加 pPieTable += 3;

HardwareSetup.c

```
118 Parameters: None
119 Returns: None
120 ************************************************************************/
121 void PieVectTable_Initialize(void)
122 {
123     int16 i;
124     PINT *pPieTable = (void *) &PieVectTable;  // Pointer to PIE vector table
125
126     EALLOW;  // Enable writes to protected registers
127     pPieTable += 3;
128     // Initialize all 128 interrupt vectors to the default reserved ISR
129     for(i = 0; i < 128; i++)
130     {
131         *pPieTable++ = rsvd_ISR;
132     }
133
134     EDIS;  // Disable writes to protected registers
135
136     // Enable the PIE Vector Table
137     PieCtrlRegs.PIECTRL.bit.ENPIE = 1;
138 }
```

1.3 更改保留函数

搬运到 RAM 执行，添加看门狗复位，改成如下。

HardwareSetup.c

```
336 Returns: None
337 *****************************************************************************/
338 int gErrorIntCnt;  // Global counter for unexpected interrupt occurrences
339 #pragma CODE_SECTION(rsvd_ISR, "ramfuncs");
340 interrupt void rsvd_ISR(void)
341 {
342     gErrorIntCnt++;  // Monitor and count unexpected interrupt entries
343     EALLOW;
344     SysCtrlRegs.WDCR= 0x0028;
345     EDIS;
346     while(1);
347 }
348
349 /*****************************************************************************
```

1.4 确认软件配置 Flash 擦除时间及更新 FlashAPI

确认软件配置 Flash 擦除时间（100M），下图为推荐配置

```
    EALLOW;  // Enable writes to protected registers
    Flash_CPUScaleFactor = SCALE_FACTOR;  // Set CPU scaling factor for Flash timing
    Flash_CallbackPtr = 0;                // Initialize callback pointer
    EDIS;   // Disable writes to protected registers
```

```
66 ------------------------------------------------------------------------------*/
67 #define CPU_RATE    10.000L   // for a 100MHz CPU clock speed (SYSCLKOUT)
68 //#define CPU_RATE   16.667L   // for a 60MHz CPU clock speed (SYSCLKOUT)
69 //#define CPU_RATE   20.000L   // for a 50MHz CPU clock speed  (SYSCLKOUT)
70 //#define CPU_RATE   25.000L   // for a 40MHz CPU clock speed  (SYSCLKOUT)
71 //#define CPU_RATE   33.333L   // for a 30MHz CPU clock speed  (SYSCLKOUT)
72 //#define CPU_RATE   41.667L   // for a 24MHz CPU clock speed  (SYSCLKOUT)
73 //#define CPU_RATE   50.000L   // for a 20MHz CPU clock speed  (SYSCLKOUT)
74 //#define CPU_RATE   66.667L   // for a 15MHz CPU clock speed  (SYSCLKOUT)
75 //#define CPU_RATE  100.000L   // for a 10MHz CPU clock speed  (SYSCLKOUT)
76
77 //---------------------------------------------------------------------------
78
79
80 //---------------------------------------------------------------------------
81 // **** DO NOT modify the code below this line ****
82 //---------------------------------------------------------------------------
83 #define SCALE_FACTOR  1048576.0L*( (200L/CPU_RATE) )  // IQ20
```

更新 API 程序，版本如下：

ADP32F036_027_Flash_API_AUTOCLK.lib

2 APP 程序

2.1 更改 Flash 等待时间

100M 主频，Flash 等待时间 3，3，5 改成 4，4，5。

HardwareSetup.c

```
192// This function MUST be executed out of RAM. Executing it
193// out of OTP/Flash will yield unpredictable results
194//---------------------------------------------------------------------
195void InitFlash(void)
196{
197   EALLOW;
198
199   //Enable Flash Pipeline mode to improve performance of code ex
200   //FlashRegs.FOPT.bit.ENPIPE = 1;
201   FlashRegs.FOPT.all = 0x01;
202   //Set the Random Waitstate for the Flash
203   FlashRegs.FBANKWAIT.bit.RANDWAIT = READ_RAND_FLASH_WAITE;
204   //Set the Paged Waitstate for the Flash
205   FlashRegs.FBANKWAIT.bit.PAGEWAIT = READ_PAGE_FLASH_WAITE;
206   //Set the Waitstate for the OTP
207   FlashRegs.FOTPWAIT.bit.OTPWAIT = READ_OTP_WAITE;
208
209   //Set number of cycles to transition from sleep to standby
210   //FlashRegs.FSTDBYWAIT.bit.STDBYWAIT = 0x01FF;
211   //Set number of cycles to transition from standby to active
212   //FlashRegs.FACTIVEWAIT.bit.ACTIVEWAIT = 0x01FF;
213   EDIS;
214
215   asm(" RPT #7 || NOP");
216}
```

2.2 更改中断向量

跳过地址 0xD00~0xD05，0xD00~xD05 不属于中断向量，添加 pPieTable += 3;

HardwareSetup.c

```
436
437//---------------------------------------------------------------------
438//初始化的时候首先把所有的中断服务程序 都初始化为默认服务程序
439//---------------------------------------------------------------------
440void InitPieVectTable(void)
441{
442    int16   i;
443    PINT *pPieTable = (void *) &PieVectTable;
444    pPieTable += 3;
445    EALLOW;
446    for(i=0; i < 128; i++)
447    {
448        *pPieTable++ = rsvd_ISR;
449    }
450    EDIS;
451
452    // Enable the PIE Vector Table
453    PieCtrlRegs.PIECTRL.bit.ENPIE = 1;
454}
```

2.3 更改保留函数

搬运到 RAM 执行，添加看门狗复位，改成如下

HardwareSetup.c

```
420 /*****************************************************************
421 所有误操作的中断处理
422 *****************************************************************/
423 int16 gErrorIntCnt;
424 #pragma CODE_SECTION(rsvd_ISR, "ramfuncs");
425 interrupt void rsvd_ISR(void)
426 {
427     gErrorIntCnt++;
428     EALLOW;
429     SysCtrlRegs.WDCR= 0x0028;
430     EDIS;
431     while(1);
432 }
433
```

2.4 休眠函数参考例程

①在 LINEN 拉低前一定要确保 DSP 封波（即 EPWWM2A/2B,PWM3A/3B,PWM4A/4B 内连引脚全部置低）；

②休眠时序要满足 LIN_EN 拉低 3us 以内，拉低 LIN_TXD 且低电平维持到 DSP 关闭；

③让芯片休眠后完全掉电（VDDIO 完全掉电到 0 V）后再进行唤醒操作（在此之前 LIN 总线不能再发数据），避免休眠异常唤醒状态。如果 LIN 总线上由于受到干扰导致休眠失败，可以增加软件逻辑功能。

HardwareSetup.c

```
1665
1666 #pragma CODE_SECTION(EnterSleepMode, "ramfuncs");
1667 void EnterSleepMode(void)
1668 {
1669     // 注意：一般逻辑是接收到休眠唤醒命令后要休眠，进来此函数之前已经要停机！
1670     // 确保外面没有停机此处再加上！
1671     DisableDrive();
1672
1673     DINT;
1674
1675     LIN_EN_LOW();    /* LEN_EN = 0; */
1676
1677     asm(" RPT #50 || NOP");
1678
1679     EALLOW;
1680     GpioCtrlRegs.GPAMUX1.bit.GPIO14 = 0;
1681     GpioCtrlRegs.GPADIR.bit.GPIO14 = 1;
1682     GpioDataRegs.GPACLEAR.bit.GPIO14 = 1;   /* TX = 0;  */
1683     EDIS;
1684
1685     ADP32F03x_usDelay(20000L);              /* 1ms    */
1686
1687     EALLOW;
1688     SysCtrlRegs.WDCR = 0x0000;
1689     EDIS;
1690
1691     while(1);
1692 }
```

2.5 确认软件配置 Flash 擦除时间及更新 FlashAPI

如 APP 程序有操作 Flash 的逻辑，需确认 Flash 擦除时间配置，如下图

```
EALLOW;  // Enable writes to protected registers
Flash_CPUScaleFactor = SCALE_FACTOR;  // Set CPU scaling factor for Flash timing
Flash_CallbackPtr = 0;                // Initialize callback pointer
EDIS;    // Disable writes to protected registers
```

```
66 ------------------------------------------------------------------------------*/
67 #define CPU_RATE    10.000L   // for a 100MHz CPU clock speed (SYSCLKOUT)
68 //#define CPU_RATE   16.667L   // for a 60MHz CPU clock speed (SYSCLKOUT)
69 //#define CPU_RATE   20.000L   // for a 50MHz CPU clock speed  (SYSCLKOUT)
70 //#define CPU_RATE   25.000L   // for a 40MHz CPU clock speed  (SYSCLKOUT)
71 //#define CPU_RATE   33.333L   // for a 30MHz CPU clock speed  (SYSCLKOUT)
72 //#define CPU_RATE   41.667L   // for a 24MHz CPU clock speed  (SYSCLKOUT)
73 //#define CPU_RATE   50.000L   // for a 20MHz CPU clock speed  (SYSCLKOUT)
74 //#define CPU_RATE   66.667L   // for a 15MHz CPU clock speed  (SYSCLKOUT)
75 //#define CPU_RATE  100.000L   // for a 10MHz CPU clock speed  (SYSCLKOUT)
76
77 //---------------------------------------------------------------------------
78
79
80 //---------------------------------------------------------------------------
81 // **** DO NOT modify the code below this line ****
82 //---------------------------------------------------------------------------
83 #define SCALE_FACTOR  1048576.0L*( (200L/CPU_RATE) )  // IQ20
```

更新 API 程序，版本如下：

